

Tutorial Test 1-1

Tuesday Morning

25 July 2017

1. Parsecs (abbreviation is pc) is a unit of length used in astronomy to measure enormous interstellar distances. 1 parsec is approximately 3.26 light-years, and a light year is the distance light travels in 1 year (365 days). If the speed of light is 3.00×10^8 m/s, how many meters are there in 1 apc (attoparsec)? Show clearly all the conversions.
Parsecs (afkorting pc) is die eenheid van lengte wat in astronomie gebruik word om enorme interstellêre afstande te meet. 1 pc is ongeveer 3.26 lig-jaar, en 'n lig-jaar is die afstand wat lig in 1 jaar (365 dae) beweeg. Indien die spoed van lig 3.00×10^8 m/s, hoeveel meter is daar in 1 apc (attoparsec)? Wys duidelik alle omskakelings.
2. Determine the area between the t -axis and the graph of $v = 3 + 4t$ which is limited by $t = 1$ and $t = 5$. The units for v are meters per second and that of t is seconds. What are the units of the area calculated?
Bereken die oppervlak tussen die t -as en die grafiek van $v = 3 + 4t$ wat begrens word deur $t = 1$ en $t = 5$. Die eenhede van v is meter per sekonde en die van t is sekonde. Wat is die eenhede van die oppervlakte wat bereken is?

Tutorial Test 1-1

1. $1 \text{ pc} = 3.26 \text{ light-years (ly)}$

$1 \text{ light-year} = 3 \times 10^8 \text{ m/s} \times 1 \text{ year.}$

$$1 \text{ apc} = 1 \times 10^{-18} \text{ pc} \times \frac{3.26 \text{ ly}}{1 \text{ pc}} \times \frac{3 \times 10^8 \text{ m/s} \times 1 \text{ yr}}{1 \text{ ly}} \times \frac{365 \text{ days}}{1 \text{ yr}} \times \frac{24 \text{ h}}{1 \text{ day}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{60 \text{ s}}{1 \text{ min}}$$

$$= 0.0308 \text{ m}$$

$$= 3.08 \times 10^{-2} \text{ m}$$

2. Area under the graph of a function
 $\Rightarrow$ Integrate.

$$V = 3 + 4t$$

$$\begin{aligned} \text{Area} &= \int_1^5 V dt = \int_1^5 (3 + 4t) dt \\ &= [3t + 2t^2]_1^5 \\ &= [(3)(5) + 2(5)^2 - (3)(1) + 2(1)^2] \\ &= 65 - 5 \\ &= 60 \end{aligned}$$

Units of the area under the graph: m (meters)

Tutorial Test 1-2

Tuesday Morning

25 July 2017

1. A nanocentury is a unit sometimes used in computing. The term is believed to have been coined by IBM in 1969 from the design objective "never to let the user wait more than a few nanocenturies for a response". If you have to wait for 1 nanocentury, how many seconds would you wait? Clearly show all the conversions.
'n Nano-eeu is 'n eenheid wat partykeer in die rekenaarbedryf gebruik word, het blykbaar by IBM begin in 1969 vanuit die ontwerp beginsel "never to let the user wait more than a few nanocenturies for a response". Indien jy vir 1 nano-eeu moet wag, hoeveel sekondes moet jy wag?

2. If $y = \frac{x^2}{\sqrt{4-x^2}}$, determine the gradient of the function at $x = 1$ and $x = -1$.

Indien $y = \frac{x^2}{\sqrt{4-x^2}}$, bepaal die gradient van die funksie by $x = 1$ and $x = -1$.

Tutorial Test 1-2

$$\begin{aligned} 1.1 \text{ nanocentury} &= 1 \times 10^{-9} \text{ century} \times \frac{100 \text{ yr}}{1 \text{ cent.}} \times \frac{365 \text{ days}}{1 \text{ yr}} \\ &\quad \times \frac{24 \text{ h}}{1 \text{ day}} \times \frac{60 \text{ min}}{1 \text{ h}} \times \frac{60 \text{ s}}{1 \text{ min}} \\ &= 3.15 \text{ s} \\ &= 3.2 \text{ s} \checkmark \end{aligned}$$

$$2. \quad y = \frac{x^2}{\sqrt{4-x^2}}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[x^2 (4-x^2)^{-\frac{1}{2}} \right] \quad \text{Using the product rule} \\ &= 2x (4-x^2)^{-\frac{1}{2}} + x^2 \left(-\frac{1}{2} (4-x^2)^{-\frac{3}{2}} (-2x) \right) \\ &= \frac{2x}{\sqrt{4-x^2}} + (-2x^3) \left(\frac{1}{2} (4-x^2)^{-\frac{3}{2}} \right) \end{aligned}$$

$$\begin{aligned} \text{At } x = 1: \quad \frac{dy}{dx} &= 1.155 + (-0.192) \\ &= 0.96 \checkmark \end{aligned}$$

$$\begin{aligned} \text{At } x = -1: \quad \frac{dy}{dx} &= 1.155 + 0.192 \\ &= 1.35 \checkmark \end{aligned}$$

One can also use the quotient rule.

Tutorial Test 1-3

Tuesday Afternoon

25 July 2017

1. Parsecs (abbreviation is pc) is a unit of length used in astronomy to measure enormous interstellar distances. 1 parsec is approximately 3.26 light-years, and a light year is the distance light travels in 1 year (365 days). If the speed of light is 3.00×10^8 m/s, how many femtoparsecs (fpc) are there in 15.0 cm? Show clearly all the conversions.

Parsecs (afkorting pc) is die eenheid van lengte wat in astronomie gebruik word om enorme interstellêre afstande te meet. 1 pc is ongeveer 3.26 lig-jaar, en 'n lig-jaar is die afstand wat lig in 1 jaar (365 dae) beweeg. Indien die spoed van lig 3.00×10^8 m/s, hoeveel femtoparsecs (fpc) is daar in 15.0 cm? Wys duidelik alle omskakelings.

2. If $y = (3x^2 + 5x)(2x^3 - 3)$, determine the gradient of the function at $x = 2$ and $x = 0$.
Indien $y = (3x^2 + 5x)(2x^3 - 3)$, bepaal die gradient van die funksie by $x = 2$ en $x = 0$.

Tutorial Test 1-3

1. $1 \text{ femto parsec} = 1 \times 10^{-15} \text{ pc} \times \frac{3.26 \text{ ly}}{1 \text{ pc}} \times \frac{3 \times 10^8 \text{ m/s} \times \text{lyr}}{1 \text{ yr}} \times \frac{365 \text{ days}}{1 \text{ yr}} \times \frac{24 \text{ hr}}{1 \text{ day}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{60 \text{ s}}{1 \text{ min}}$

$$= 30.8 \text{ m}$$

$$15 \text{ cm} = 15 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}} \times \frac{1 \text{ fpc}}{30.8 \text{ m}}$$
$$= 4.86 \times 10^{-3} \text{ fpc}$$

2. $y = (3x^2 + 5x)(2x^3 - 3)$

$$\frac{dy}{dx} = (6x + 5)(2x^3 - 3) + (3x^2 + 5x)(6x^2)$$

$$\text{At } x = 2: \frac{dy}{dx} = 221 + 528 = 749$$

$$\text{At } x = 0: \frac{dy}{dx} = -15$$

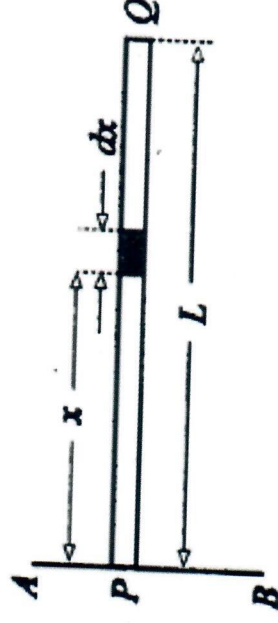
Max: 10

1. A unit sometimes used in computing, the term is believed to have been coined by IBM in 1969 from the design objective "never to let the user wait more than a few nanocenturies for a response". If you have to wait for 54 s, how many nanocenturies would you wait? Clearly show all the conversions.

'n Nano-eeu is 'n eenheid wat partykeer in die rekenaarbedryf gebruik word, het blykbaar by IBM begin in 1969 vanuit die ontwerp beginsel "never to let the user wait more than a few nanocenturies for a response". Indien jy vir 54 s moet wag, hoeveel nano-eeue moet jy wag?

2. The sketch shows a thin straight homogeneous rod with length L metres and mass M kilograms. The mass per unit length is given by $\mu = M/L \text{ kg m}^{-1}$. AB is an axis perpendicular to the rod, PQ , and dx is an infinitesimal element of length at position x from AB . The mass of this element is given by $dm = \mu dx = (M/L)dx$. The quantity $dI = x^2 dm$ is known as the moment of inertia of the element about the axis AB . Calculate the moment of inertia, I , of the entire rod about axis AB .

Die skets toon 'n dun reguit homogene staaf met massa M kilogram en lengte L meter. Die massa per eenheidslengte word gegee deur $\mu = M/L \text{ kg m}^{-1}$. AB is 'n as wat loodreg is op PQ en dx is 'n infinitesimale lengte-element van die staaf op afstand x vanaf AB . Die massa van hierdie element word nou gegee deur $dm = \mu dx = (M/L)dx$. Die hoeveelheid $dI = x^2 dm$ staan bekend as die traagheidsmoment van die element dx om die as AB . Bereken nou die traagheidsmoment, I , van die hele staaf om AB .



Tutorial Test 1-4

$$\begin{aligned} 1. \quad 54 \text{ s} &= 54 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1 \text{ h}}{60 \text{ min}} \times \frac{1 \text{ day}}{24 \text{ h}} \times \frac{1 \text{ y}}{365 \text{ days}} \times \frac{1 \text{ cent.}}{100 \text{ y}} \\ &= 1.71 \times 10^{-8} \text{ centuries} \\ &= 17.1 \text{ nanocentury} \end{aligned}$$

$$2. \quad \mu = \frac{M}{L} \text{ kg} \cdot \text{m}^{-1} \qquad dm = \mu dx = \frac{M}{L} dx$$

$$dI = x^2 dm = x^2 \left(\frac{M}{L} \right) dx$$

$$I = \int dI = \int_0^L x^2 \left(\frac{M}{L} \right) dx$$

$$= \frac{M}{L} \int_0^L x^2 dx$$

$$= \frac{M}{L} \left[\frac{1}{3} x^3 \right]_0^L$$

$$= \frac{1}{3} \frac{M}{L} \cdot L^3 - 0$$

$$= \frac{1}{3} ML^2$$

Max 10
Marks.

Tutorial Test 1-5

Wednesday Afternoon

26 July 2017

1. A Warhol is a unit of fame or hype, derived from Andy Warhol's dictum "everyone will be world-famous for fifteen minutes". It represents, naturally, fifteen minutes of fame. When a student graduates, he/she is on the stage for about 30 s. How many Warhols does this amount to. Give your answer in terms of three significant figures, using the correct prefix for your units.

'n Warhol is 'n eenheid vir beroemdheid wat deur Andy Warhol se gesegde "everyone will be world-famous for fifteen minutes" sy oorsprong het. Dit verteenwoordig, natuurlik, vyftien minute se roem. Wanneer 'n student graad vang, is sy/hy vir omtrent 30 s op die verhoog. Hoeveel Warhols is hier ter sprake. Gee jou antwoord in terme van drie beduidende syfers en gebruik die korrekte voorvoegsel vir jou eenhede.

1. A projectile is fired across a field with an initial velocity of 40 m/s. The horizontal position from the launch point is given by x and the vertical position (height) by y . The equation which describes the motion is given by $y = -\frac{x^2}{20} + \frac{x}{\sqrt{3}}$, where x and y are both in meters. For what value of x will the projectile reach its maximum height? What is this max. height?
'n Projektiel word oor 'n horisontale veld afgeskiet teen 'n begin spoed van 40 m/s. Die horisontale posisie vanaf die lanseerpunt word gegee deur x , en die vertikale posisie (hoogte) deur y . Die vergekyking van die baan wat die projektiel volg word gegee deur $y = -\frac{x^2}{20} + \frac{x}{\sqrt{3}}$, waarin beide x en y in meter gemeet word. Vir watter waarde van x bereik die projektiel sy maksimum hoogte? Wat is die maks. hoogte?

Tutorial Test 1-5

1.

$$30 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1 \text{ Warhol}}{15 \text{ min}} = 0.0333 \text{ Warhols}$$
$$= \underline{33.3 \text{ m Warhols}}$$

3 significant
figures

small m for
"milli"

2. $y = -\frac{x^2}{20} + \frac{x}{\sqrt{3}}$

To determine minimums and maximums
in a function

$$\frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{2x}{20} + \frac{1}{\sqrt{3}} = 0$$

$$2x = \frac{1}{\sqrt{3}} \cdot 20$$

$$x = \frac{10}{\sqrt{3}} \approx 5.77 \text{ m}$$
$$= 5.8 \text{ m} \checkmark$$

$$y_{\max} = -\frac{x^2}{20} + \frac{x}{\sqrt{3}}$$

$$= -\frac{(5.8)^2}{20} + \frac{(5.8)}{\sqrt{3}} = 1.66 \text{ m}$$

$$= 1.7 \text{ m} \checkmark$$

2. Tutorial Test 1-6

Wednesday Afternoon 26 July 2017

2. A Warhol is a unit of fame or hype, derived from Andy Warhol's dictum "everyone will be world-famous for fifteen minutes". It represents, naturally, fifteen minutes of fame. Akani Simbine's personal best time for the 100 m is 9.89 s. How many Warhols does this amount to. Give your answer in terms of three significant figures, using the correct prefix for your units. *'n Warhol is 'n eenheid vir beroemdheid wat deur Andy Warhol se gesegde "everyone will be world-famous for fifteen minutes" sy oorsprong het. Dit verteenwoordig, natuurlik, vyftien minute se roem. Akani Simbine se persoonlike beste tyd vir die 100 m naelloop is 9.89 s. Hoeveel Warhols is hier ter sprake tydens hierdie naelloop. Gee jou antwoord in terme van drie beduidende syfers en gebruik die korrekte voorvoegsel vir jou eenhede.*
3. A projectile is fired across a field with an initial velocity of 40 m/s. The horizontal position from the launch point is given by x and the vertical position (height) by y . The equation which describes the motion is given by $y = -\frac{x^2}{20} + \frac{x}{\sqrt{3}}$, where x and y are both in meters. For what value of x will the projectile reach its maximum height? What is this max. height? *'n Projektiel word oor 'n horisontale veld afgeskiet teen 'n begin spoed van 40 m/s. Die horisontale posisie vanaf die lanseerpunt word gegee deur x , en die vertikale posisie (hoogte) deur y . Die vergekyking van die baan wat die projektiel volg word gegee deur $y = -\frac{x^2}{20} + \frac{x}{\sqrt{3}}$, waarin beide x en y in meter gemeet word. Vir watter waarde van x bereik die projektiel sy maksimum hoogte? Wat is die maks. hoogte?*

Tutorial Test 1-6

1.

$$9.895 \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1 \text{ Warhol}}{15 \text{ min}} = 0.01098 \text{ Warhols}$$

$$= \underline{11.0 \text{ m Warhols}}$$

3 significant
figures.

small m for
"milli"

2.

Same as 1-5 question 2.